

As an election official, I want to be able to run a Single Transferable Vote (STV) election as requested by the Secretary of State.

Acceptance Criteria:

- The STV election should produce the correct output so that future STV elections can be conducted correctly and fairly.
- Output a correct list of winners and losers in order of the election progress.
- Ties should be broken fairly.
- The election summary should be shown on the screen at the end of the election, and an audit file for inspection is generated.

Definition of Done:

- Refactor the code.
- Fix the STV code that has the known bug on the buglist, which occurs when there are more seats than candidates and there is only one vote, some candidates are showing up on both the winner and loser lists.
- Pass all unit/system tests.
- Complete the corresponding testing log and documentation.
- Able to compile and run on CSE lab machine

Effort: Medium, 3-4 hours.

PBI Author: Mark Tsai, Andy Dang, Michael Dunn, Manan Chaturvedi.

As an election official, I want the system to read the .csv file with newly added headers that bring in election information so that there is no room for maliciousness, possibility of fraud, and user error with manual input.

Acceptance Criteria:

- The system should be able to read the .csv file headers.
- Type of voting, number of seats, number of candidates, and number of ballots can be put in the .csv file in order as headers that the system can read directly, rather than using manual input.
- The rest of the information after the header, candidate names and ballots, as per the original file format, should be read correctly.
- The file should not be edited, and the file permissions should not be changed after the system reads it.

Definition of Done:

- Refactor the code.
- Create a function to accept multiple files (assuming there is no error and the format is correct in the file).
- Change the system to call the updated version of the function.
- Pass all unit/system tests.
- Complete the corresponding testing log and documentation.
- Able to compile and run on CSE lab machine

Effort: Medium, 3-4 hours.

PBI Author: Mark Tsai.

As an election official, I want to be able to run a municipal voting (MV) election so that voters can vote for multiple people at once for elections as requested by the Secretary of State.

Acceptance Criteria:

- The MV election should produce the correct output so that future MV elections can be conducted correctly and fairly.
- Ties should be broken fairly.
- The election summary should be shown on the screen at the end of the election, and an audit file for inspection is generated.

Definition of Done:

- Create the Municipal class, and have it inherit from Election.
- Be able to intake .csv files that are formatted for MV elections.
- Use polymorphism to use the runElection method, and have it correctly determine the winner of an MV election based on the ballots that it receives.
- Create a method to display the results of an MV election on the screen.
- Refactor the code and add functionality to the voting system to let it take in MV election ballots.
- Pass all unit/system tests.
- Complete the corresponding testing log and documentation.
- Able to compile and run on CSE lab machine

Effort: Large, 4-5 hours.

PBI Author: Andy Dang

As an election official, I want to input multiple .csv files containing ballot information for all kinds of elections so that I can collect ballots from multiple voting locations and determine election winners/losers from the collective ballot information. This is so that the election official can utilize a more accessible method of processing separate files rather than only being able to process one file.

Acceptance Criteria:

- Multiple .csv files can be input into the system and the correct candidates are selected to be winners and losers based on the collective ballot information over all ballot information.

Definition of Done:

- Refactor the code.
- Create a function for reading in multiple files (assuming there is no error and the format is correct in the file).
- Pass all unit/system tests.
- Complete the corresponding testing log and documentation.
- Able to compile and run on CSE lab machine

Effort: Small, 1-2 hours.

PBI Author: Manan Chaturvedi, Mark Tsai.

As an election official, I want to make sure that each incoming ballot running a STV algorithm is validated as requested by the Secretary of State.

Acceptance Criteria:

- At least half of the candidates should be ranked for a ballot to be valid.
- All removed ballots must be included in the audit file.
- The total number of ballots must be adjusted, subtracting from the number of invalid ballots detected.

Definition of Done:

- Refactor the code.
- Implement the function in the STV algorithm.
- Pass all unit/system tests.
- Complete the corresponding testing log and documentation.
- Able to compile and run on CSE lab machine

Effort: Medium, 3-4 hours.

PBI Author: Manan Chaturvedi, Mark Tsai.

As an election official, I want a small report to be run after each election for all types of elections (STV, MV, Plurality) that summarizes their outcomes. This is so that all elections have their respective critical information available for referring back to validate the election and ensure an integrous election is run.

Acceptance Criteria:

- There exists a small report for each election containing valid information about the election's total number of seats and ballots, candidate names and the total number of votes/percentage of total votes, and the list of winners.

Definition of Done:

- Refactor the code.
- Implement genSmallReport for MV, STV, and Plurality classes.
- Pass all unit/system tests.
- Integrate into the system with system testing functionality.
- Complete the corresponding testing log and documentation.
- Able to compile and run on CSE lab machine

Effort: Small, 1-2 hours.

PBI Author: Manan Chaturvedi.